Comptia N+

N+ NETWORK FUNDAMENTALS AND BUILDING NETWORKS

IP ADDRESSING AND SUBNETTING

TASKS:1

List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

RESULT

Class	**IP Address Range**	**Real-World Example**
Class A	1.0.0.0 – 126.255.255.255	Large companies like Google
Class B	128.0.0.0 – 191.255.255.255	University or college network
Class C	192.0.0.0 – 223.255.255.255	Home Wi-Fi router
Class D	224.0.0.0 – 239.255.255.255	Live video streaming (IPL streaming)
Class E	240.0.0.0 – 255.255.255.255	Research and testing

TASKS:2

Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Show your working steps for each calculation.

RESULT

GIVEN:

- IP Address = 192.168.10.45
- Subnet Mask = 255.255.255.0 (/24)

- **Step 1: Network Address**

  The last part becomes 0.

  Network Address = 192.168.10.0

- **Step 2: Broadcast Address**

  The last part becomes 255.

  Broadcast Address = 192.168.10.255

- **Step 3: Valid Host Range**

  First usable IP:192.168.10.1

  Last usable IP:192.168.10.254

Item	Answer
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
First Host	192.168.10.1
Last Host	192.168.10.254

TASKS:3

Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

RESULT

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Group	**Network Address**	**Subnet Mask**
Delivery Partners	10.0.0.0/26	255.255.255.192
Restaurants	10.0.0.64/26	255.255.255.192
Customers	10.0.0.128/26	255.255.255.192
Admins	10.0.0.192/26	255.255.255.192

TASKS:4

You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Show your calculation steps and final answers.

RESULT

GIVEN:

- Network = 172.16.0.0/16
- Required = 20 subnets

Step 1

Borrow 5 bits because:

$$2^5 = 32 \text{ subnets (enough for 20)}$$

Step 2

Original subnet mask:

$$16 = 255.255.0.0$$

New subnet mask:

$$21 = 255.255.248.0$$

Step 3

Host bits left:

$$32 - 21 = 11 \text{ bits}$$

Hosts per subnet:

$$2^{11} - 2 = 2046 \text{ hosts}$$

Item	Value
New Subnet Mask	255.255.248.0 (/21)
Number of Subnets	32
Hosts per Subnet	2046

TASKS:5

A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

RESULT

The IP address 192.168.1.300 is not valid because each part (called an octet) must be between 0 and 255. The number 300 is outside this range, so the device cannot use this IP address.

A valid IP address in the same subnet could be:

- 192.168.1.100

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- 192.168.1.150
- 192.168.1.200

These IP addresses are valid because all four numbers are between 0 and 255.